Iterative Time Complexity Examples

1.

```
algo sum(a, n){
 s = 0
 for (int i = 0; i < n; i++){
   s = s + a[i]
 }
 return s
}
```

2.

```
algo sum(a, n){
 s = 0
 for (int i = n; i > 0; i--){
   s = s + a[i]
 }
 return s
}
```

3.

```
 for (int i = 0; i < n; i++){
   s = s + a[i]
 }
 for (int j = 0; j < n; j++){
   s = s + b[j]
 }
```

4.

```
for (int i = 0; i < n; i = i + 1){
  break
}
```

5.

```
for (int i = 0; i < n; i = i + 2){
  print("1");
}
```

6.

```
for (int i = n; i > 0; i = i - 2){
  print("1");
}
```

7.

```
for (int i = 1; i < n; i = i * 2){
  print("1");
}
```

8.

```
for (int i = n; i > 1; i = i / 2){
  print("1");
}
```

9.

```
for (int i = 0; i*i < n; i++){
  print("1");
}
```

10.

```
algo add(a, b, n){
 for (int i = 0; i < n; i++){
  for (int j = 0; j < n; j++){
    c[i, j] = A[i, j] + B[i, j]
  }
```

```
}
```

11.

```
algo multiply(a, b, n){
 for (int i = 0; i < n; i++){
  for (int j = 0; j < n; j++){
    c[i, j] = 0
    for (int k = 0; k < n; k++){
     c[i, j] = A[i, k] * B[i, k] + c[i, j]
    }
  }
}
```

12.

```
for (int i = 0; i < n; i++){
   for (int j = 1; j < n; j = j*2){
        print(“1”);
   }
}
```

13.

```
for (int i = 0; i < n; i++){
   for (int j = 1; j < n; j = j*4){
        if (j % 5 == 0){
          break
        }
   }
}
```

14.

```
x = n
while x > 0:
   x - = x
```

15.

```
y = n
while y > 1:
   y = y//2
```

16.

```
i = 1
while i < n:
   i = i + i
```

17.

```
x = 0
for (i = 1; x < n; i = i + 1){
   x = x + i
}
```

18.

```
 for (int i = 0; i < n; i = i + 1){
  for (int j = 0; j < i; j++){
      print()
    }
  }
```

19.

```
for (int i = 1; i < n; i = i + 1){
  j = 1
  while (j*j <= i){
    j += 1
  }
}
```

20.

```
for (i = 1; i <=n*n - 10; i++){
   for (j = 1; j <=i; j++){
      print()
   }
}
```

21.

a.**[CO7]Calculate** the time complexity of the following function

Finding_Worst_Case(n):

```
int i,j,k,m,multi,a,b,c
for( i = n; i >= 1; i = i / 7 ){
      for( j = 1; j <= n; j = j + 3 ) {
            for( k=1; k<=40 ; k=k+1){
                  multi=a*b
            }
            for( m=n ; m>=1 ; m=m-5 ){
                  multi=multi*c
            }
      }
}
```